OPERATIONS RUNBOOK / INSTALLATION & ADMINISTRATION

Oracle Database 19c Enterprise Edition

Windows Server 2025 Standard Evaluation
End-to-end installation SOP with captured evidence, post-install checks, sample schema, RMAN backup, and Data Pump export/import.

Document	Value
Version / date	1.0 / 03 November 2026
Target host	WIN-JE797ASF0CA (lab VM)
Operating system	Windows Server 2025 Standard Evaluation, build 26100, x64
Installed Oracle Database	19c Enterprise Edition, 19.3.0.0.0
Database structure	CDB ORCLCDB; PDB ORCLPDB1
Execution evidence	27 screenshots captured during the completed installation
Document state	Historical installation verified; sample data, backup and Data Pump steps are prospective

CRITICAL SUPPORT LIMITATION Oracle 19.3 is not certified for Windows Server 2025. Oracle requires Oracle Database 19c Release Update 19.27 or later for this OS. You can install Oracle Database 19c (19.3) first and apply the Release Update (RU) later without reinstalling Oracle Database. Oracle officially supports post-installation patching using OPatch.

Validated milestones: Oracle 26ai Free removed; 19c software installed; DBCA created ORCLCDB and ORCLPDB1; listener and authenticated SQL*Plus PDB connection passed; ARCHIVELOG enabled; APP_DATA tablespace created. The later APP_USER, sample-data, RMAN and Data Pump operations have not been executed.

Recommended installation sequence

https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/install-oracle-database.html?source=:ex:nc:::rc_wwmk180119p00044:chatbotsinhr,:ex:nc:::rc_wwmk180119p00044:chatbotsinhr,:ex:nc:::rc_wwmk180119p00044:chatbotsinhr,:ex:nc:::rc_wwmk180119p00044:chatbotsinhr&utm_source=chatgpt.com

1. Purpose, scope and execution status

This standard operating procedure reconstructs the steps actually performed on the Windows Server lab machine, with the screenshots supplied in the installation session. It then provides repeatable, executable procedures for the next requested administration exercises.

Section	Status	Scope
Remove Oracle AI Database Free	COMPLETED	Clean deinstallation, residual home deletion and service checks
Install Oracle Database 19.3	COMPLETED	Software-only Enterprise Edition install; virtual Windows service account
Create ORCLCDB / ORCLPDB1	COMPLETED	DBCA advanced setup, FRA, ARCHIVELOG, listener, EM Express
Post-install verification	COMPLETED	SQL*Plus, CDB/PDB, listener, Windows services, startup registry, APP_DATA
Create sample tables and users	PROCEDURE ONLY	APP_USER owns sample table; APP_READER granted access
RMAN database backup	PROCEDURE ONLY	Whole CDB with archived logs and validation
Data Pump export / import	PROCEDURE ONLY	Export APP_USER; import into APP_IMPORT

1.1 Operational assumptions

- Standalone VM with administrative control and a stable, non-production network.
- An active Oracle installation home at C:\app\oracle\product\19.0.0\dbhome_1.
- Existing ORCLCDB CDB and ORCLPDB1 PDB; PDB open READ WRITE.
- Windows Oracle database service runs using NT SERVICE\OracleServiceORCLCDB.
- Demonstrations use SQL*Plus and RMAN installed inside Oracle Home; no separate client needed.
- All passwords are entered privately at prompts. Never include passwords in screenshots, scripts, tickets or console transcripts.
- Paths C:\OracleBackups and C:\OracleExports are examples for an isolated lab. Production backup media must be separate, protected and preferably off host.

2. Prerequisites and downloads

Item	Recorded / required value
OS	Windows Server 2025 Standard Evaluation x64; 10.0.26100
Oracle server media	V982656-01.zip — Oracle Database 19.3 Windows x64, approximately 2.9 GB
Optional separate client	V982658-01.zip was also downloaded; not used for this server installation
Installer access	Local Windows Administrator and sufficient C: disk space
Oracle Home	C:\app\oracle\product\19.0.0\dbhome_1
Oracle Base	C:\app\oracle
Listener	LISTENER / TCP 1521
FRA	C:\app\oracle\fast_recovery_area\ORCLCDB; 12,732 MB configured
Supported Windows Server 2025 release	19c RU 19.27 or later; patch unavailable during this lab build

2.1 Official distribution and documentation

- Oracle Software Delivery Cloud: https://edelivery.oracle.com/osdc/faces/SoftwareDelivery
- Oracle 19c Windows downloads: https://www.oracle.com/database/technologies/oracle19c-windows-downloads.html
- Oracle Database 19c Windows installation guide: https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/
- Oracle OS support checklist (including minimum RU): https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/operating-system-checklist-oracle-database-installation-microsoft-windows.html

- My Oracle Support (updates / OPatch, subject to entitlement): https://support.oracle.com/

LICENSE / SUPPORT An Oracle account and downloaded Enterprise Edition binaries do not establish production licensing or patch entitlement. Confirm customer license, support CSI, feature entitlements and operating-system support before production use.

3. Remove the previous Oracle 26ai Free installation

Historical action completed on this machine. The operator explicitly requested deletion without data backup. On any other server, obtain a separate data-retention decision before uninstalling.

1. Open Control Panel → Programs and Features (appwiz.cpl), select Oracle AI Database 26ai Free, then Uninstall.
2. Complete the Oracle deinstaller and restart Windows.
3. Verify no Oracle services or Oracle/TNS environment variables remain. The old 26ai directory initially contained only deinstall logs/response files; it was removed after inspection.

```
Get-CimInstance Win32_Service | Where-Object Name -Match 'Oracle' |
  Select-Object Name,State,StartMode
Get-ChildItem Env: | Where-Object Name -Match 'ORACLE|TNS'
Get-ChildItem -LiteralPath 'C:\app\oracle\26ai' -Force
# Only when all contents are confirmed residual 26ai artifacts:
Remove-Item -LiteralPath 'C:\app\oracle\26ai' -Recurse -Force
Test-Path 'C:\app\oracle\26ai'    # recorded result: False
```

DESTRUCTIVE STEP Do not repeat this section on the currently running 19c server. It is included as history and for rebuilding a fresh lab only.

4. Install the Oracle 19c Enterprise Edition software

4.1 Prepare Oracle Home and extract the image

4. Use V982656-01.zip (database server image). Do not use the separate Oracle client installer to build the server.
5. Create C:\app\oracle\product\19.0.0\dbhome_1, then extract the image contents directly into that final Oracle Home.
6. Verify setup.exe is immediately under dbhome_1; do not run setup.exe from a temporary Downloads extraction directory.

```
$home = 'C:\app\oracle\product\19.0.0\dbhome_1'
Test-Path "$home\setup.exe"              # True
& "$home\setup.exe"                      # or right-click Run as administrator
```

4.2 Oracle Universal Installer (OUI) screens

Use the following documented choices. Screenshots were captured from the actual successful installation and may show software defaults prior to correction.

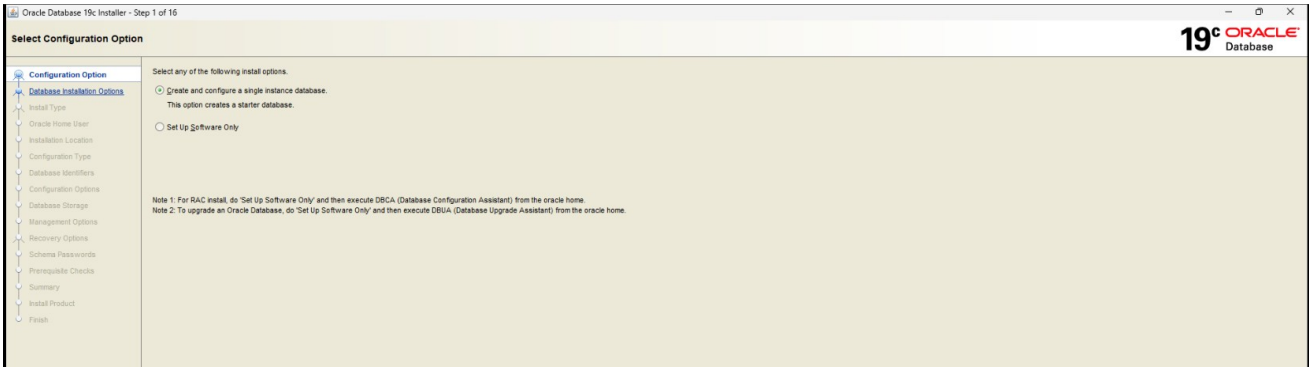


Figure 01. OUI configuration option — Set Up Software Only.



Figure 02. OUI database installation options — Single instance database installation.

7. Select Set Up Software Only; choose Single instance database installation; select Enterprise Edition.

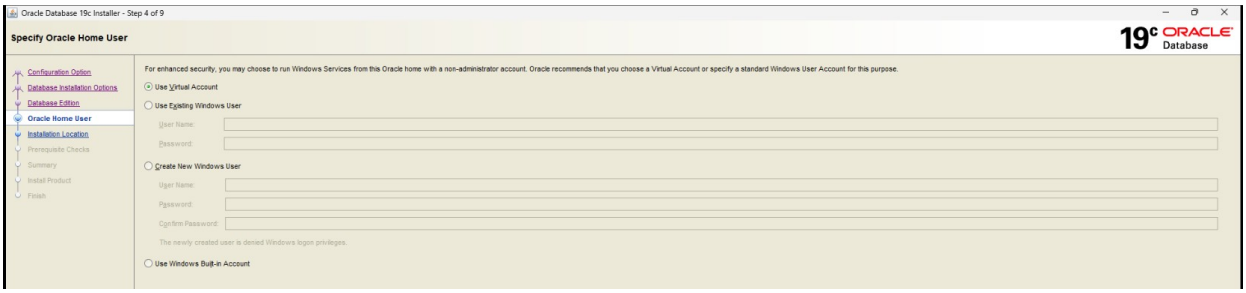


Figure 03. OUI home user — Use Virtual Account selected.

8. Use Virtual Account for the Oracle Home User; the Oracle database service later runs as a virtual service account. It also avoids running Oracle database services under the highly privileged Windows LocalSystem account.

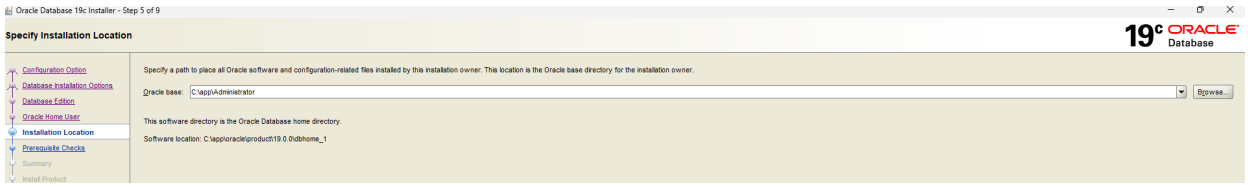


Figure 04. OUI location screen as initially shown; Oracle Base needed correction from C:\app\Administrator.

9. Set Oracle Base to C:\app\oracle and software location (Oracle Home) to C:\app\oracle\product\19.0.0\dbhome_1.

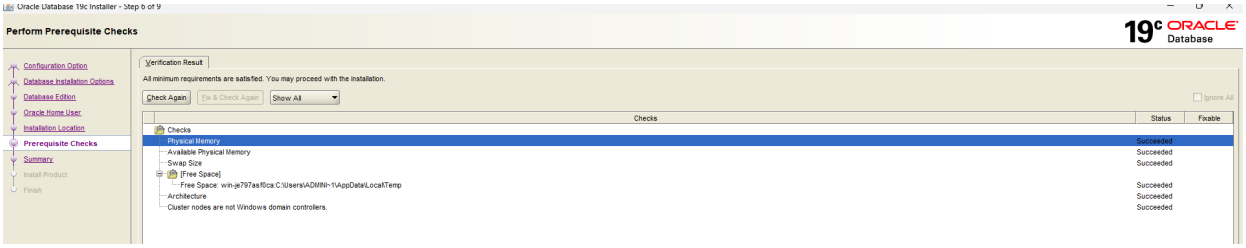


Figure 05. OUI prerequisite checks — all visible tests succeeded.

10. Review physical memory, swap, free disk space, architecture and domain-controller checks. Do not override failures. Passing the installer checks does not certify this OS/version combination.

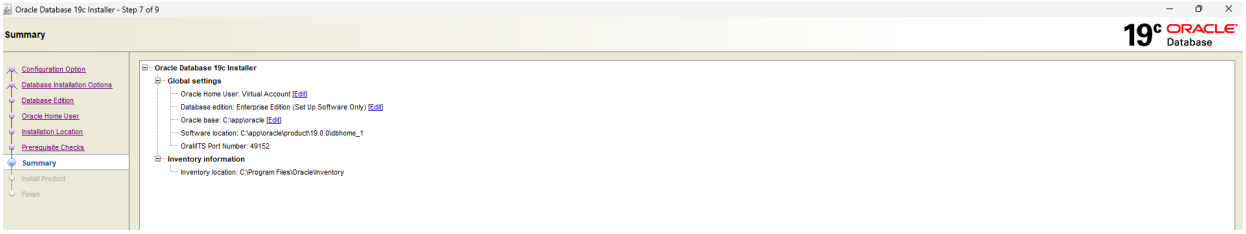


Figure 06. OUI summary with Enterprise Edition, virtual account, correct Oracle Base and Home.

11. Review summary then click Install. Do not create the database in OUI because the chosen mode is Software Only.



Figure 07. OUI finish — Oracle Database registration successful.

4.3 Verify installed tools and Oracle registry

```
$home = 'C:\app\oracle\product\19.0.0\dbhome_1'
Test-Path "$home\bin\sqlplus.exe"     # recorded True
& "$home\bin\sqlplus.exe" -v        # recorded Version 19.3.0.0.0
Test-Path "$home\bin\dbca.bat"       # recorded True
Get-ItemProperty 'HKLM:\SOFTWARE\ORACLE\KEY_OraDB19Home1' |
  Select-Object ORACLE_HOME,ORACLE_BASE,ORACLE_HOME_NAME
```

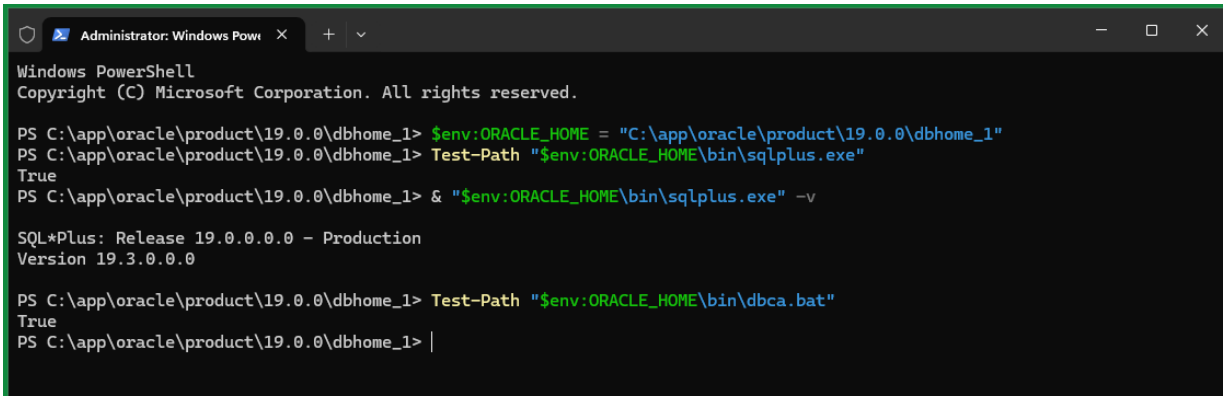


*Figure 08. PowerShell verification — SQL*Plus 19.3 available and DBCA present.*

Recorded Oracle Home: C:\app\oracle\product\19.0.0\dbhome_1; Oracle Base: C:\app\oracle; Oracle Home name: OraDB19Home1.

5. Create the CDB and PDB in DBCA

Launch Database Configuration Assistant from Start → Oracle - OraDB19Home1 → Database Configuration Assistant, or double-click C:\app\oracle\product\19.0.0\dbhome_1\bin\dbca.bat.

5.1 DBCA: create database and deployment

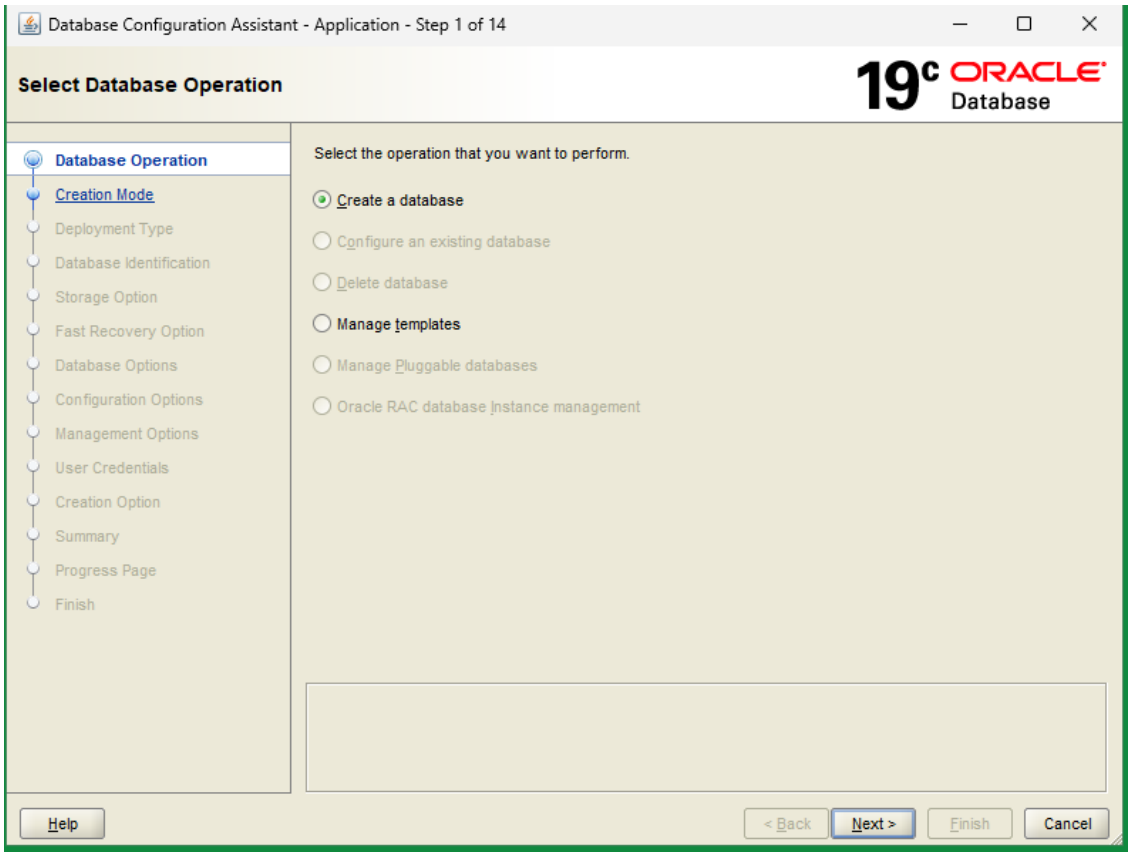


Figure 09. DBCA Step 1 — Create a database.

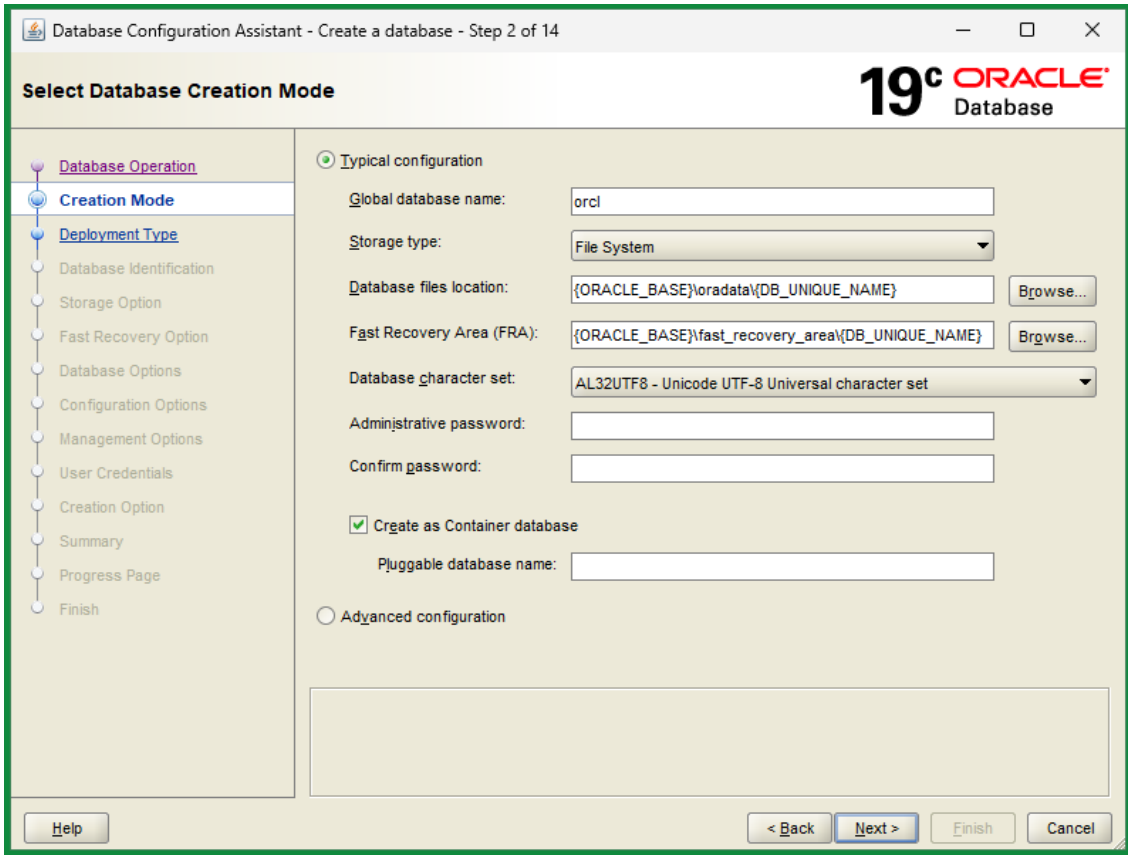


Figure 10. DBCA Step 2 — select Advanced configuration (not typical).

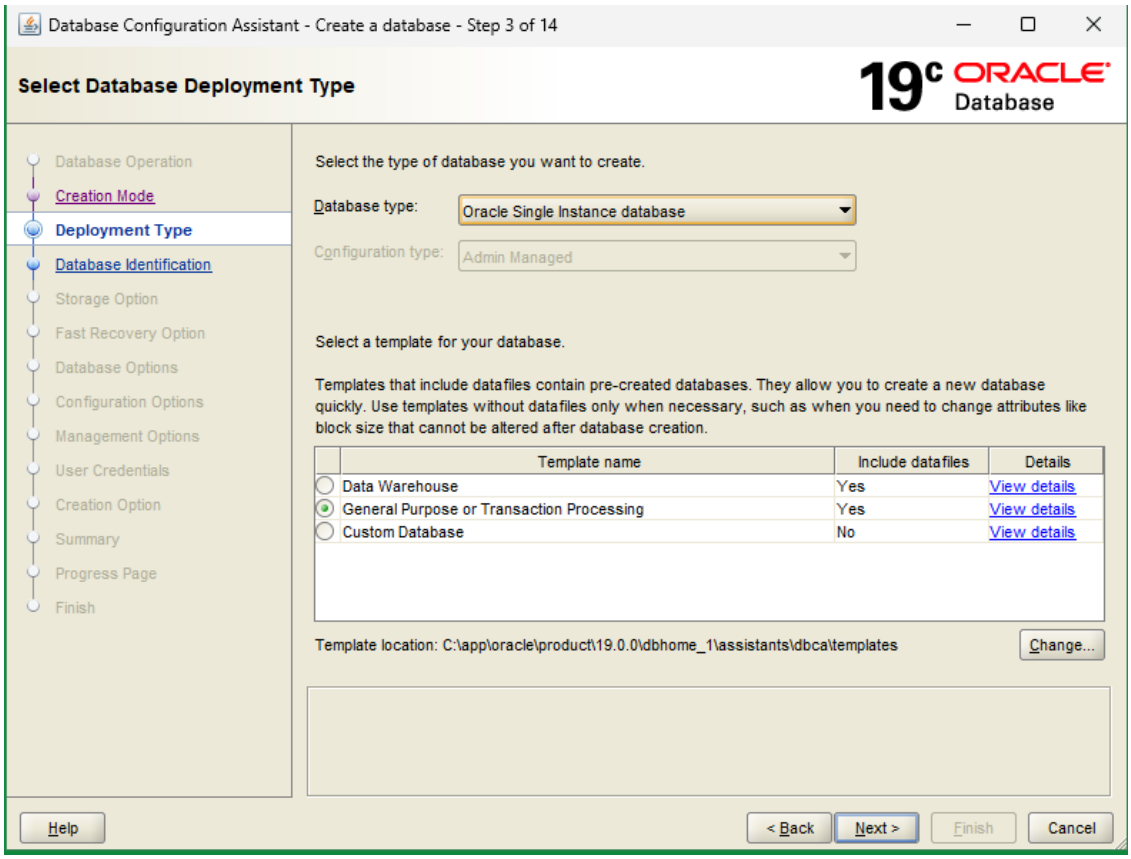


Figure 11. DBCA Step 3 — Oracle Single Instance database; General Purpose or Transaction Processing.

12. Select Create a database → Advanced configuration → Oracle Single Instance database → General Purpose or Transaction Processing template.

5.2 DBCA: database identity and storage

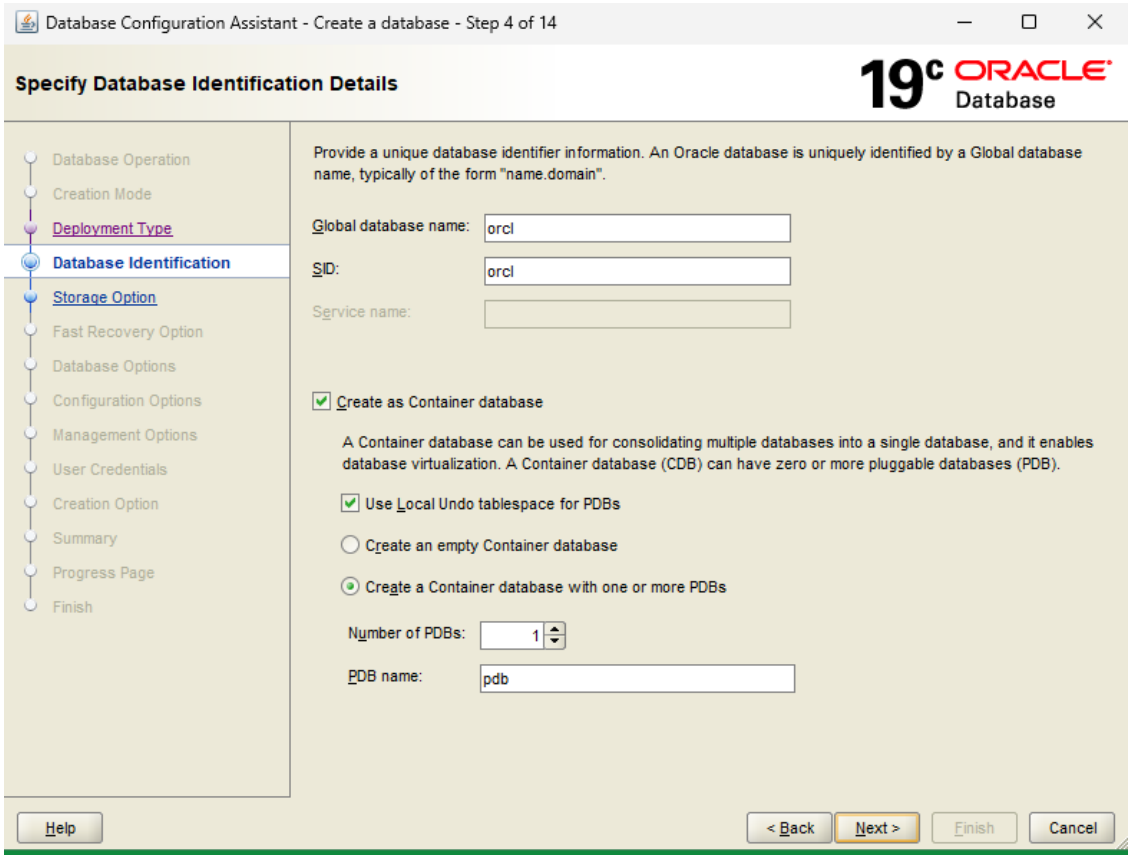
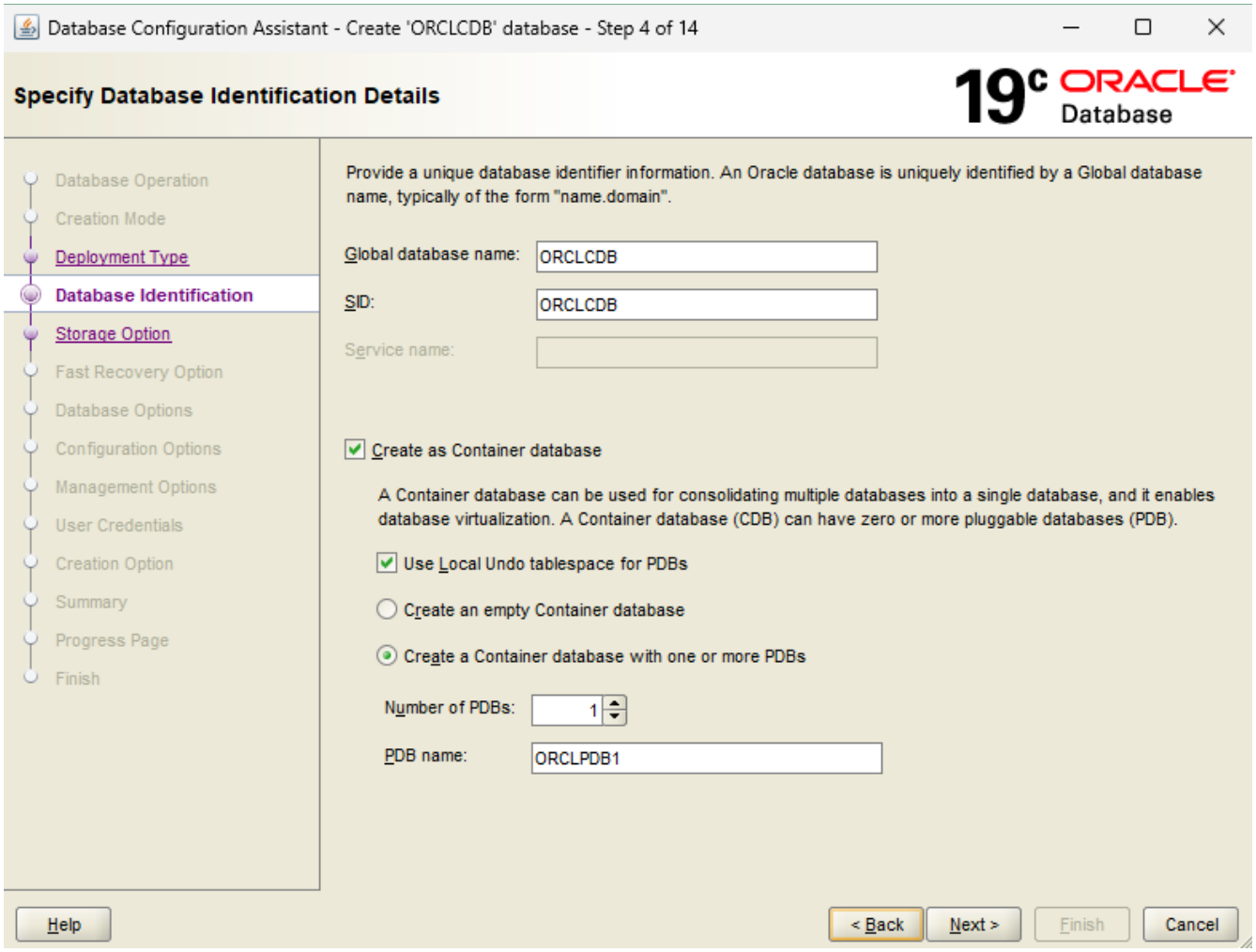


Figure 12. DBCA Step 4 — initial identification values prior to setting ORCLCDB/ORCLPDB1.

Change to below



Setting	Final configured value
Global database / SID	ORCLCDB / ORCLCDB
Create as container database	Yes
Create a PDB	Yes; one PDB named ORCLPDB1
Local Undo	Enabled

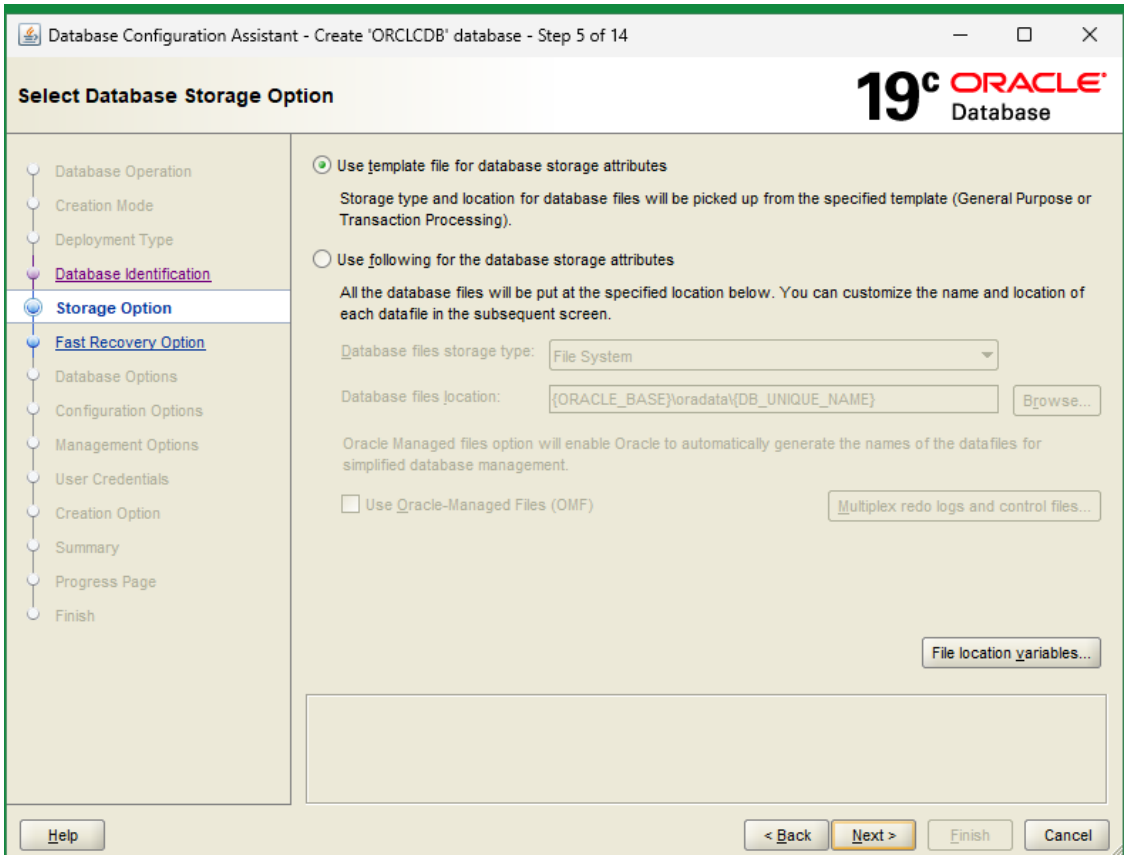


Figure 13. DBCA Step 5 — use template file for database storage attributes.

13. Keep template-defined database file-system storage in C:\app\oracle\oradata\ORCLCDB. An explicit Oracle Managed Files default was not configured.

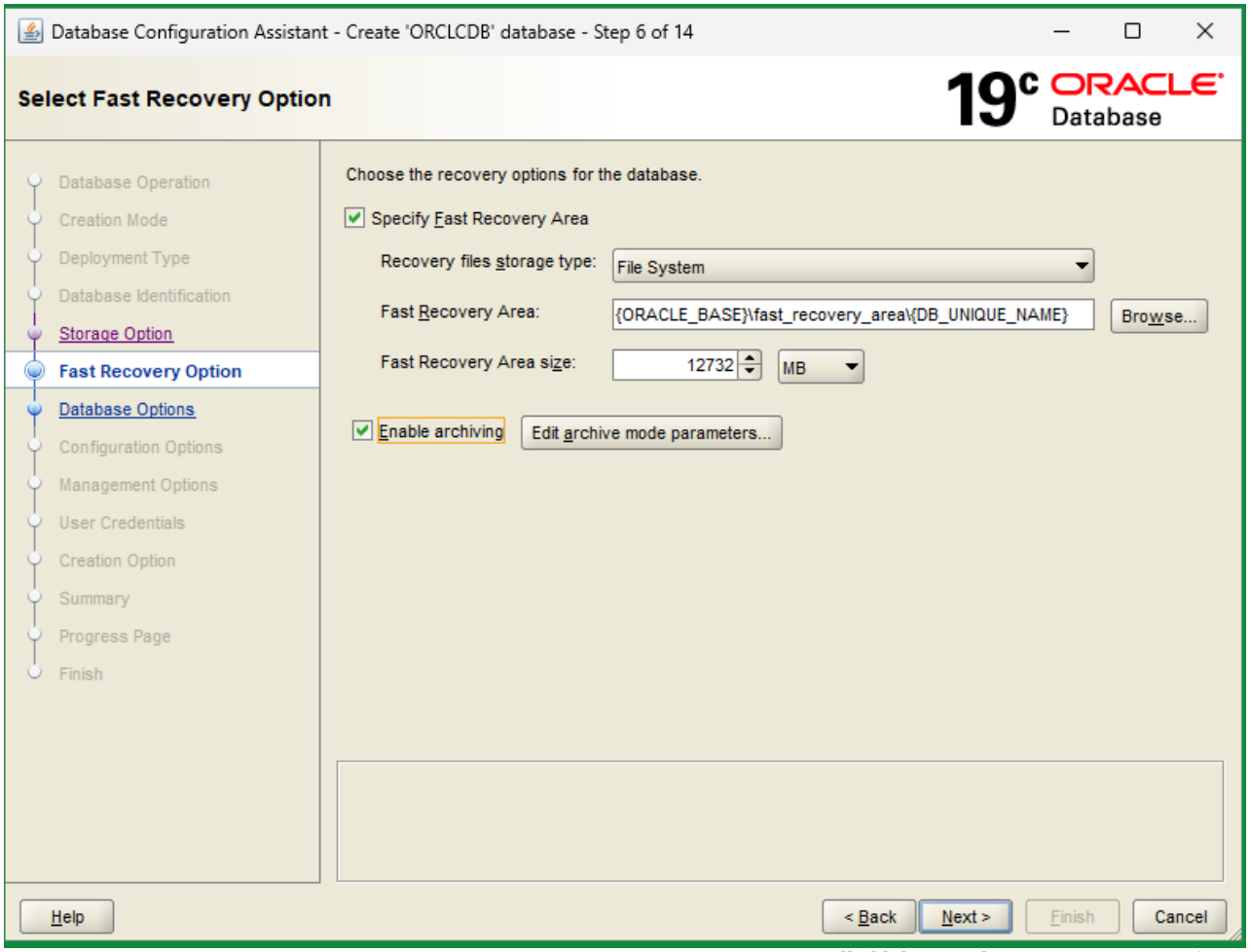


Figure 14. DBCA Step 6 — Fast Recovery Area (FRA) configured; screenshot shows archiving box prior to final selection.

14. Specify FRA, File System, default recovery area, 12,732 MB. Enable ARCHIVELOG mode. ARCHIVELOG was subsequently verified through v$database.

5.3 DBCA: listener, security and instance parameters

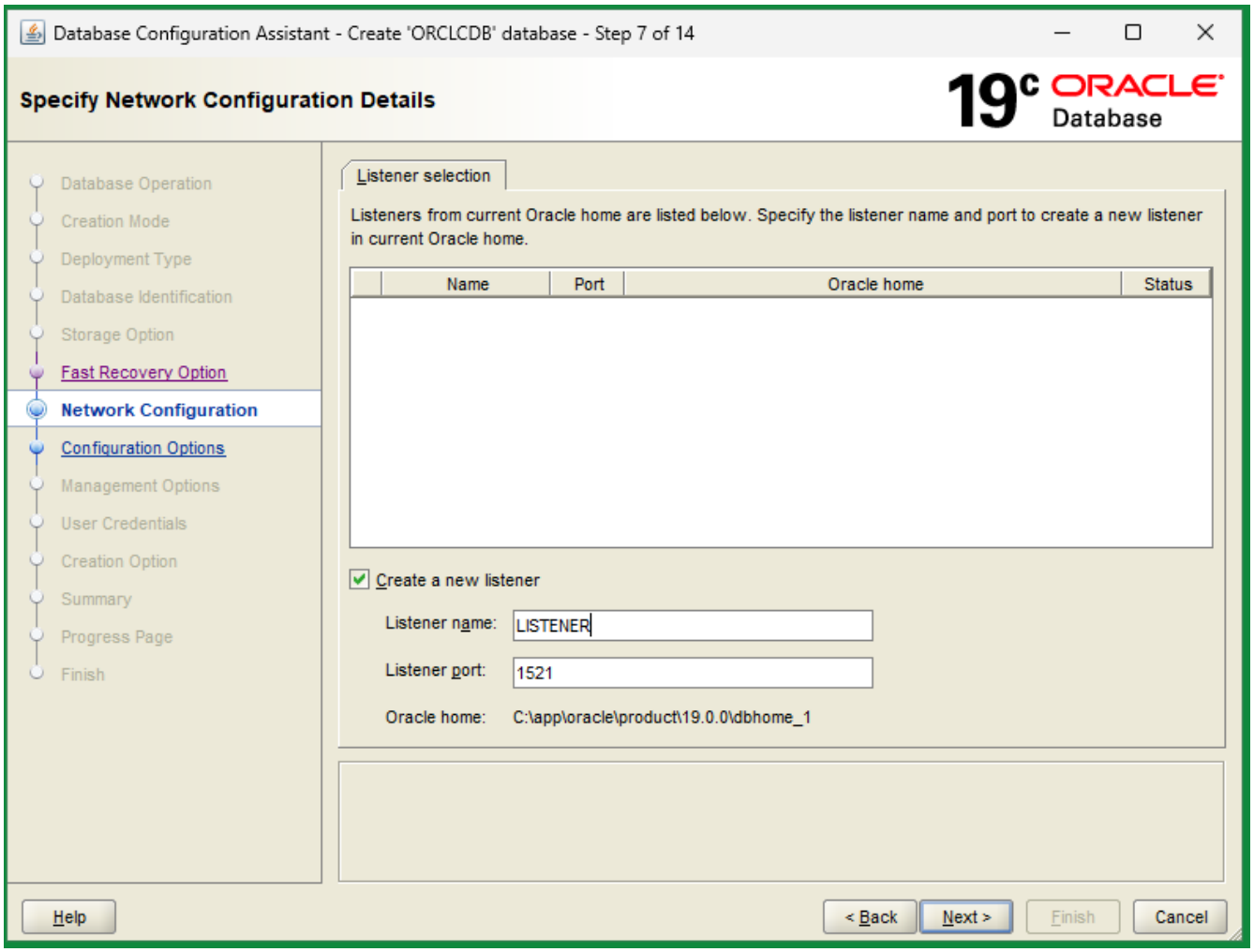


Figure 15. DBCA Step 7 — create a new listener named LISTENER on port 1521.

15. Create new listener LISTENER / TCP 1521.

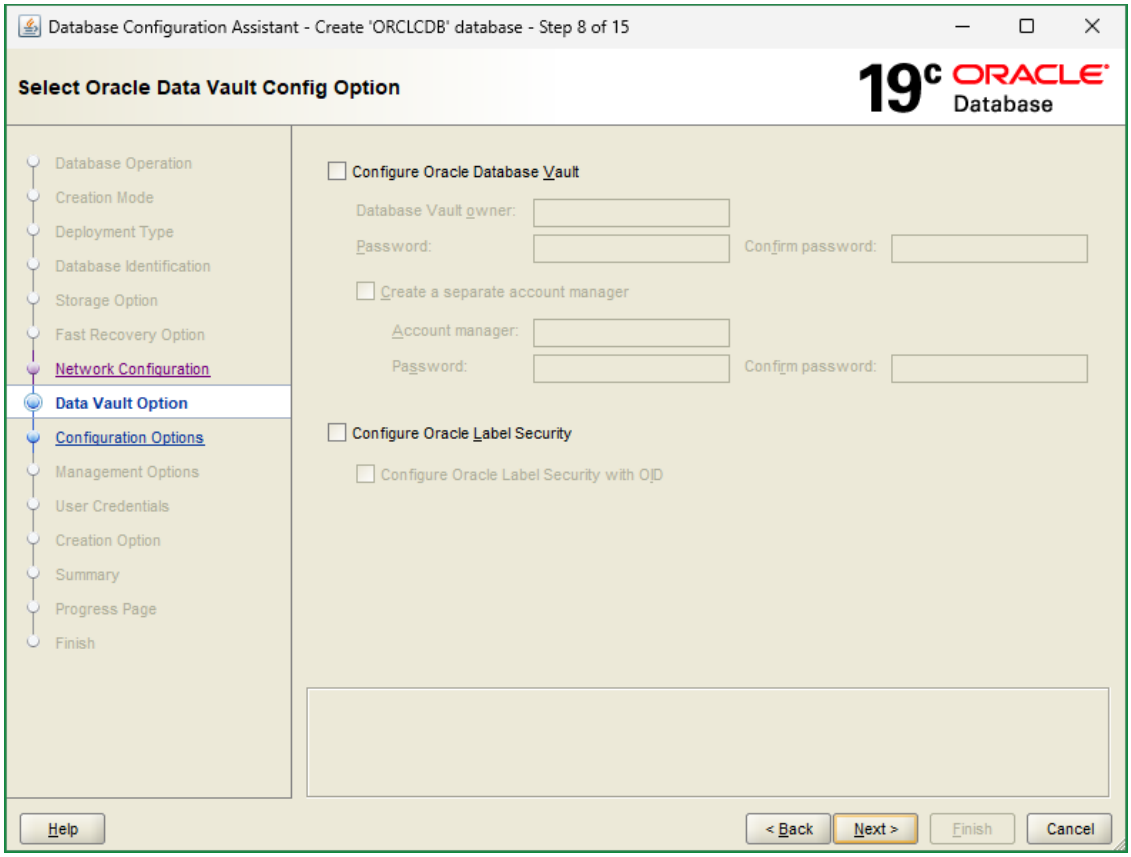


Figure 16. DBCA Step 8 — Database Vault and Label Security left disabled.

16. Leave optional Data Vault and Label Security unchecked unless separately licensed, scoped and configured.

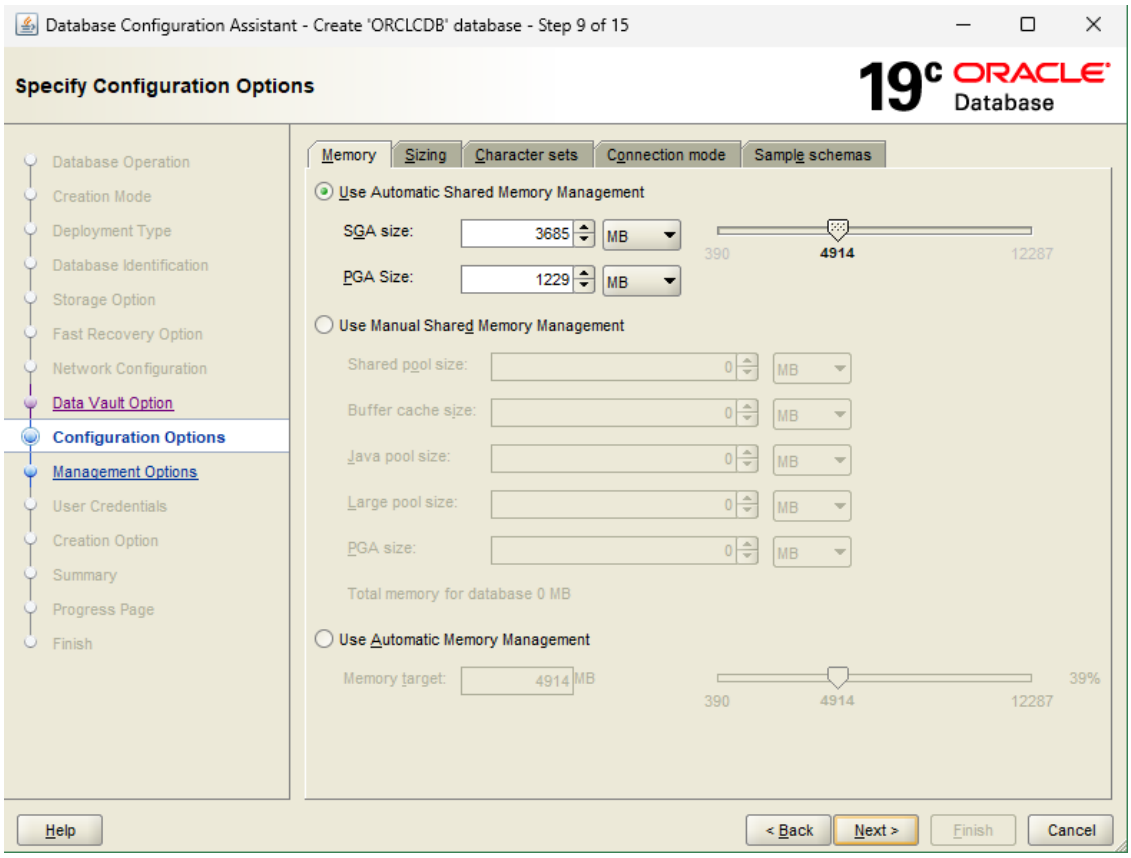


Figure 17. DBCA memory tab — ASMM with SGA 3,685 MB and PGA 1,229 MB.

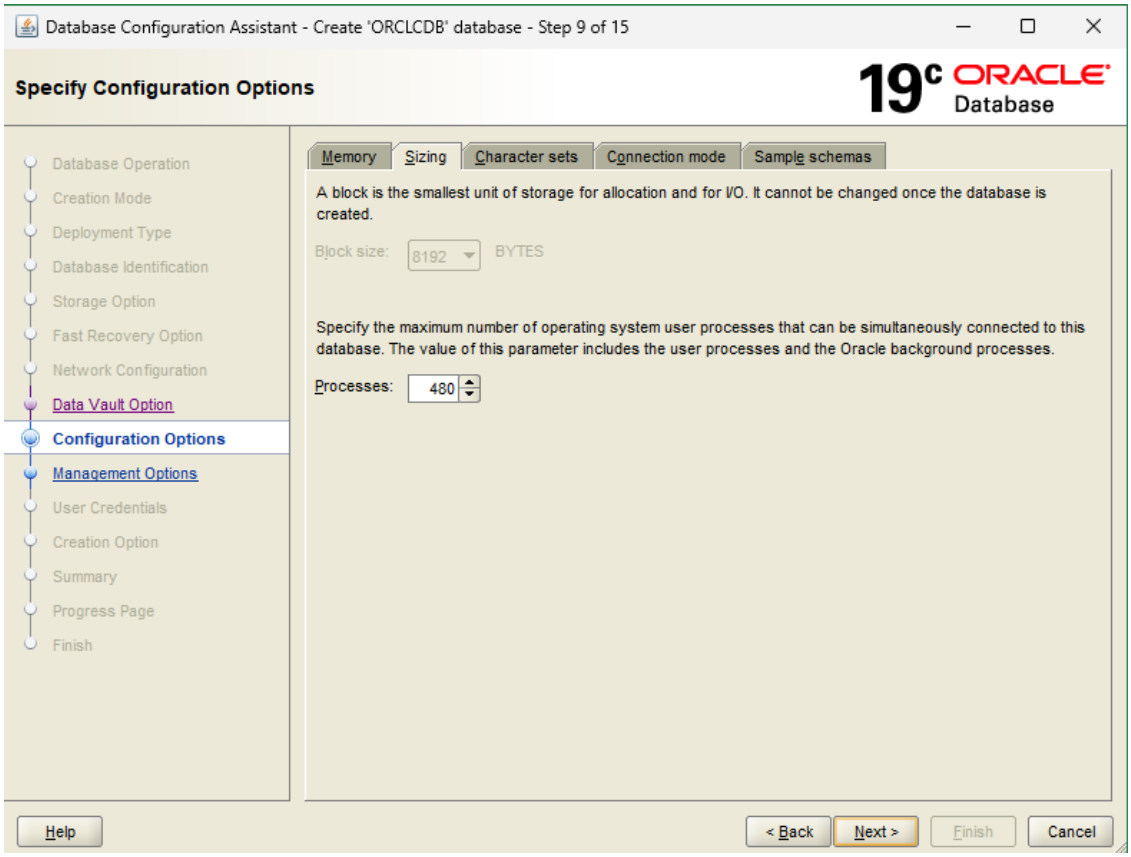


Figure 18. DBCA sizing tab — 8 KB blocks and PROCESSES=480.

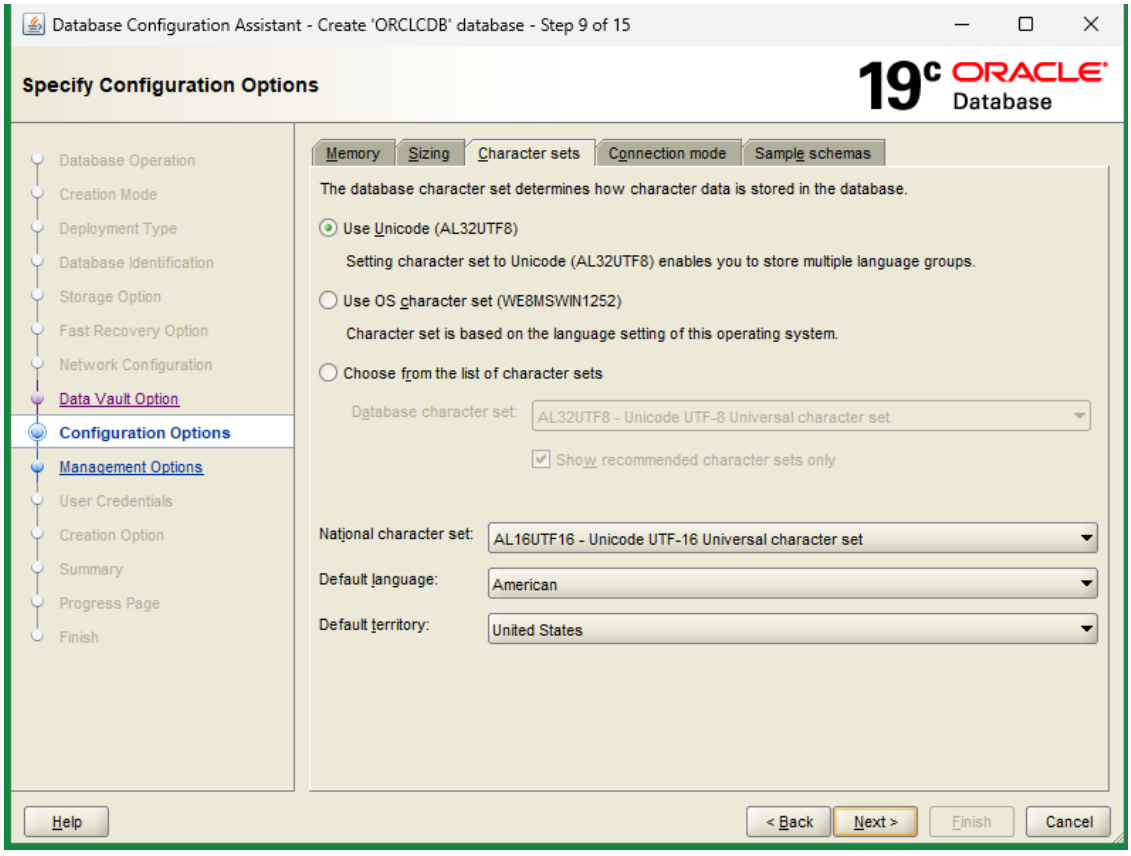


Figure 19. DBCA character sets tab — AL32UTF8 and AL16UTF16.

17. Use AL32UTF8, national character set AL16UTF16, language American and territory India (final NLS_TERRITORY=**INDIA**).

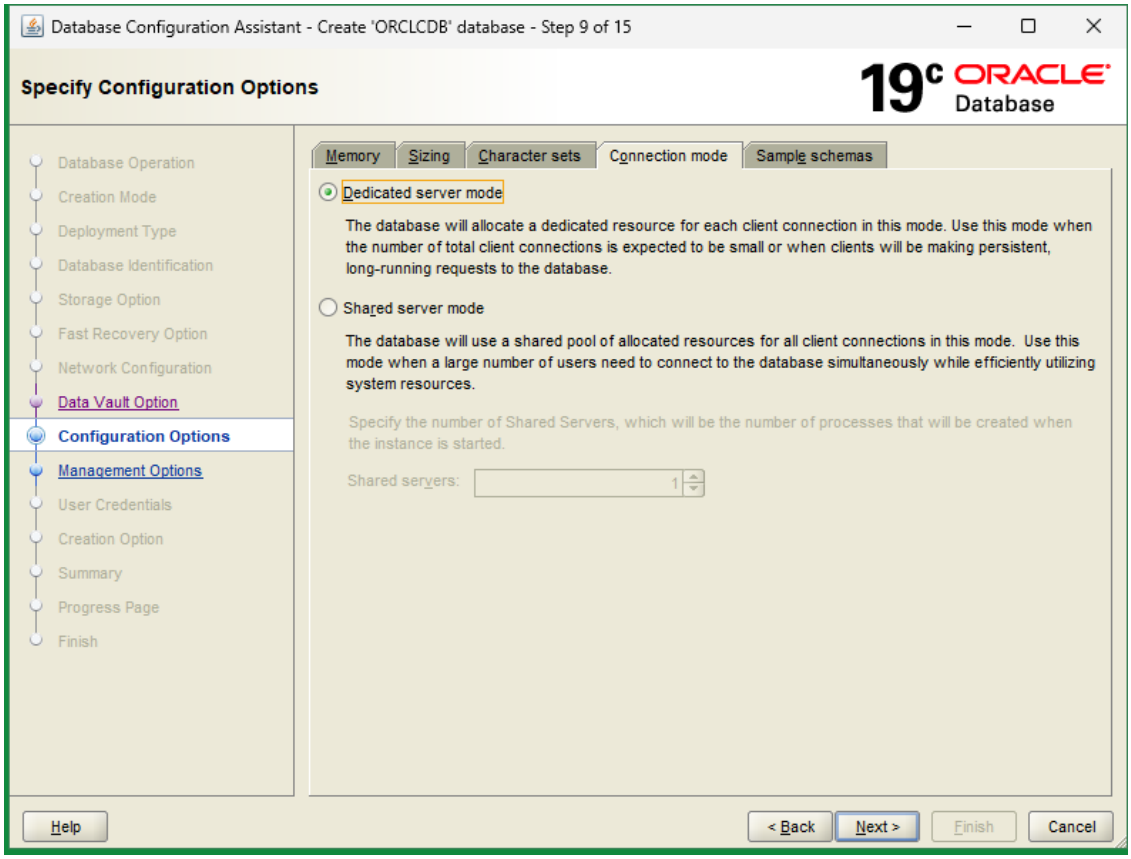


Figure 20. DBCA connection mode tab — dedicated server.

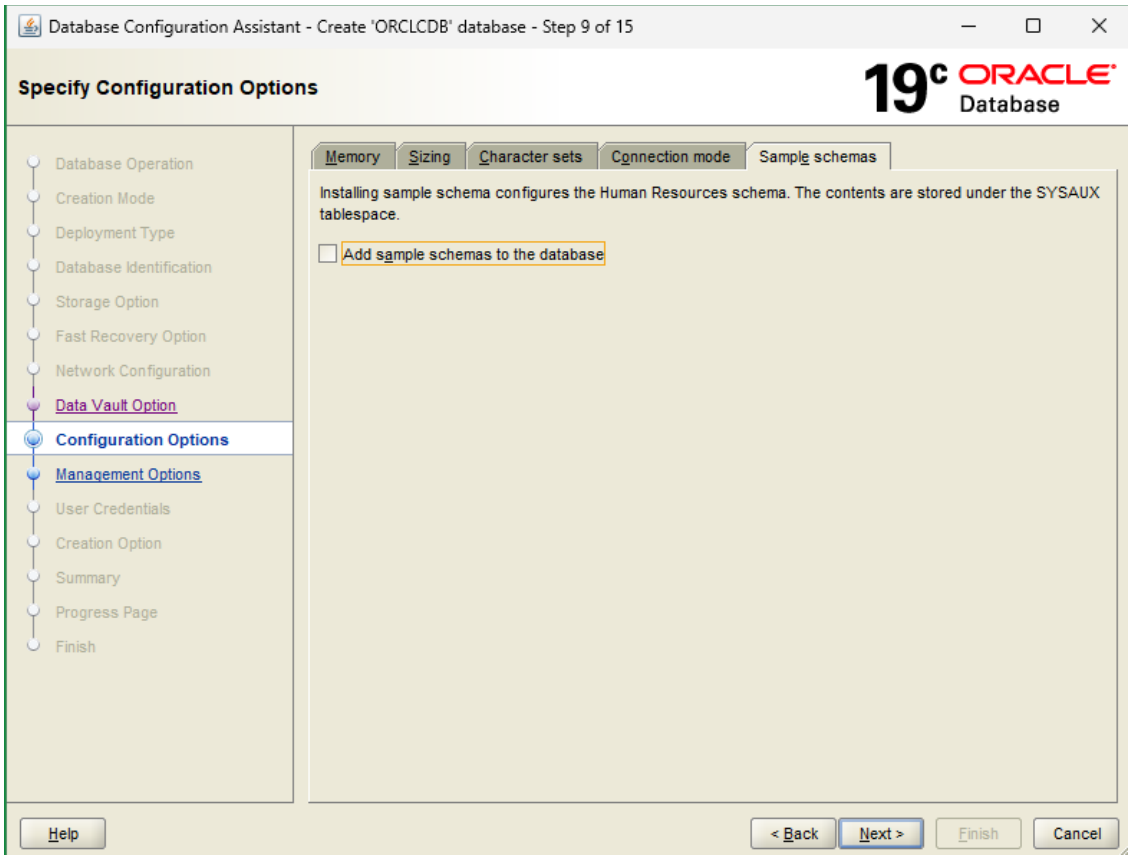


Figure 21. DBCA sample schemas tab — do not install sample schemas at DBCA stage.

18. Use Dedicated server mode; leave Add sample schemas unchecked (sample schema is created later in this SOP).

5.4 DBCA: management, passwords, creation and summary

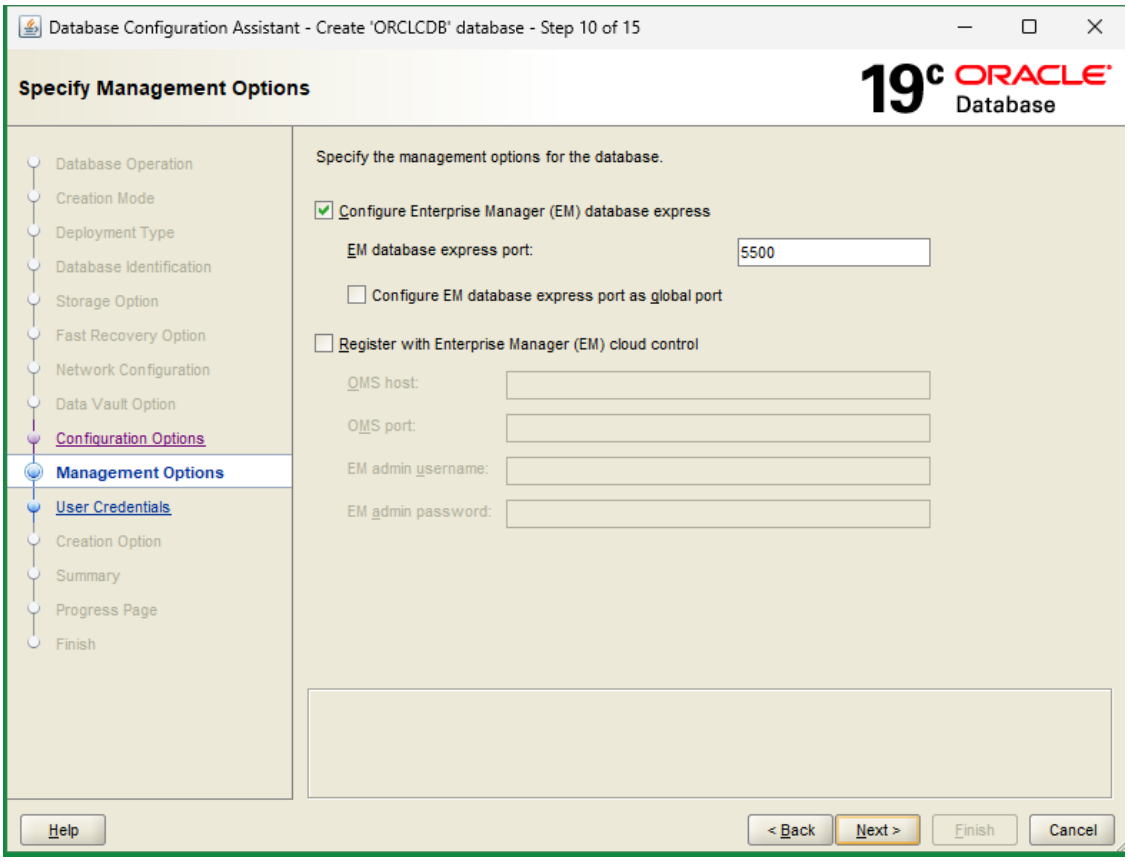


Figure 22. DBCA management screen — EM Database Express enabled on 5500.

`https://localhost:5500/em`

19. Configure Enterprise Manager Database Express on HTTPS/TCPS port 5500; do not register with EM Cloud Control.

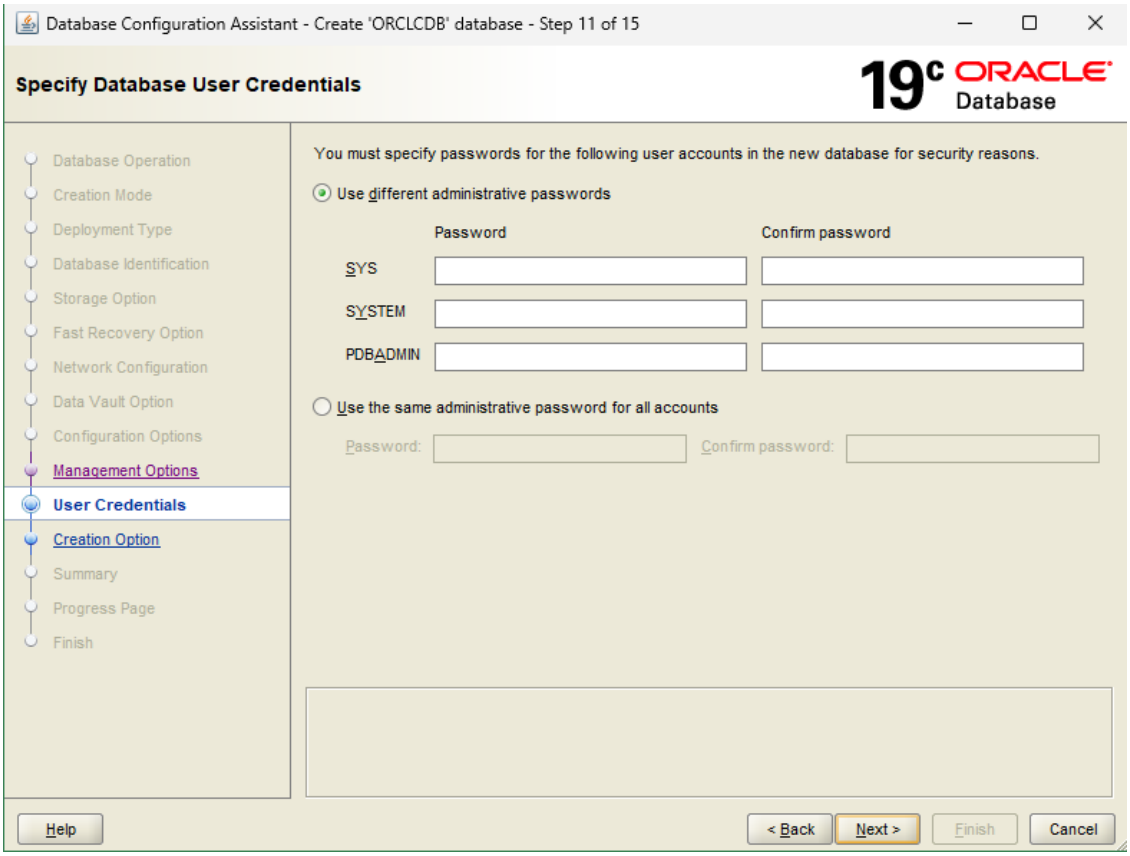
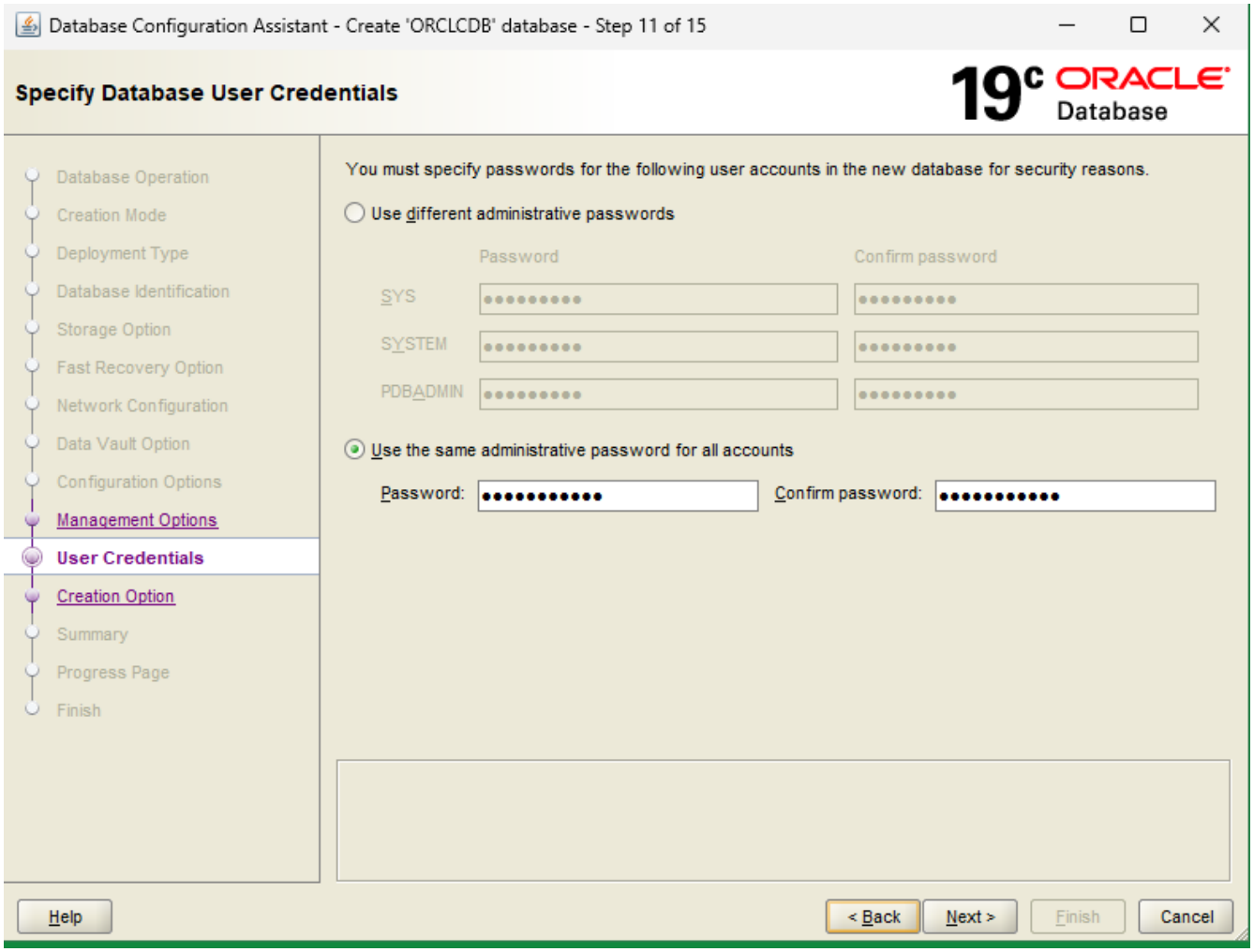


Figure 23. DBCA admin password screen — no passwords displayed.

20. Set strong credentials for SYS, SYSTEM and PDBADMIN. The lab used a common initial password; rotate to separate credentials for any operational use.



Do not use “@” in the password string

Figure 24. DBCA creation options — Create database; optionally generate creation scripts.

21. Keep Create database checked; no post-creation script. Generate scripts optionally; protect them as they may contain sensitive configuration.

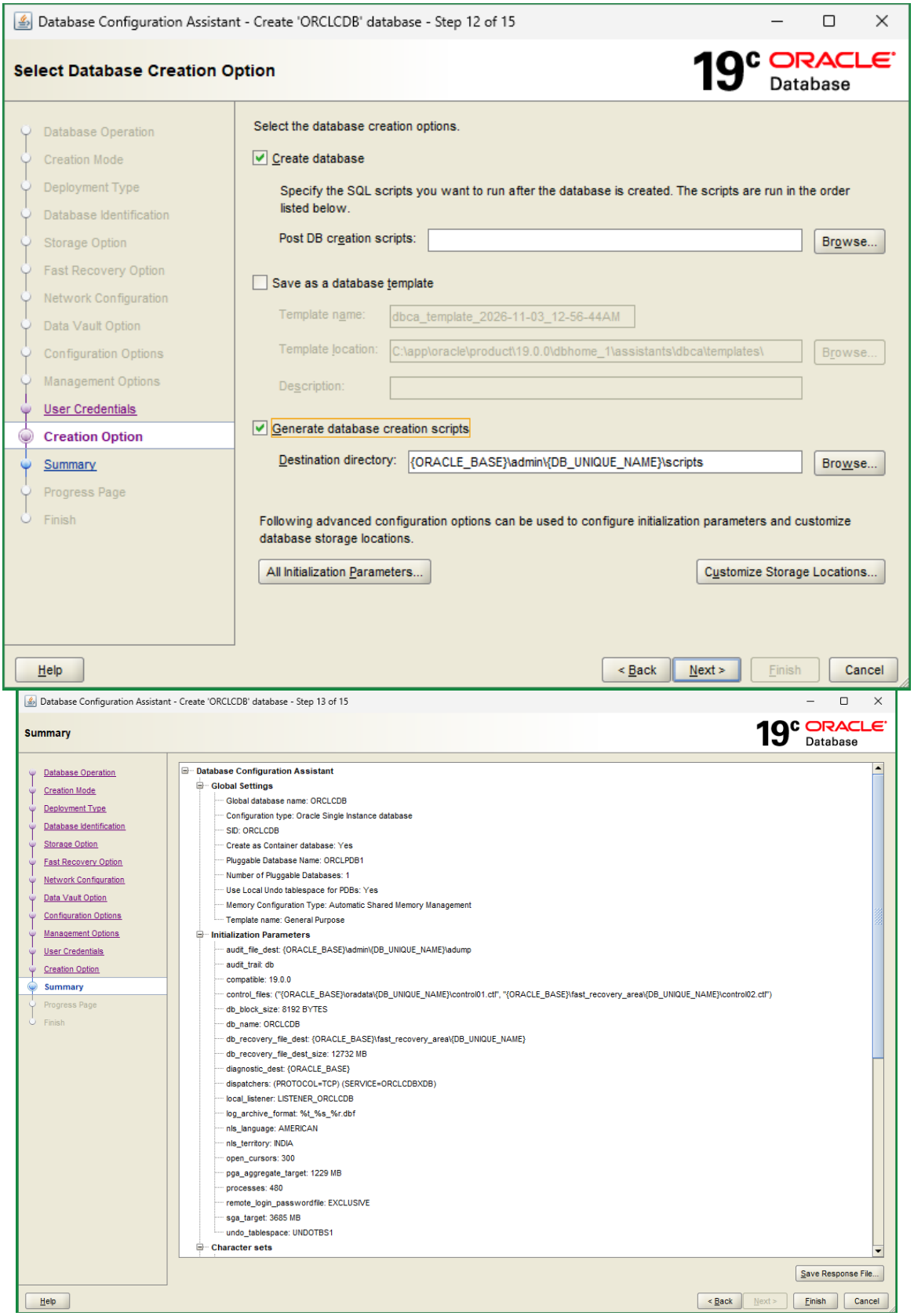


Figure 25. DBCA summary — ORCLCDB CDB, ORCLPDB1, ASMM, regional and recovery settings.

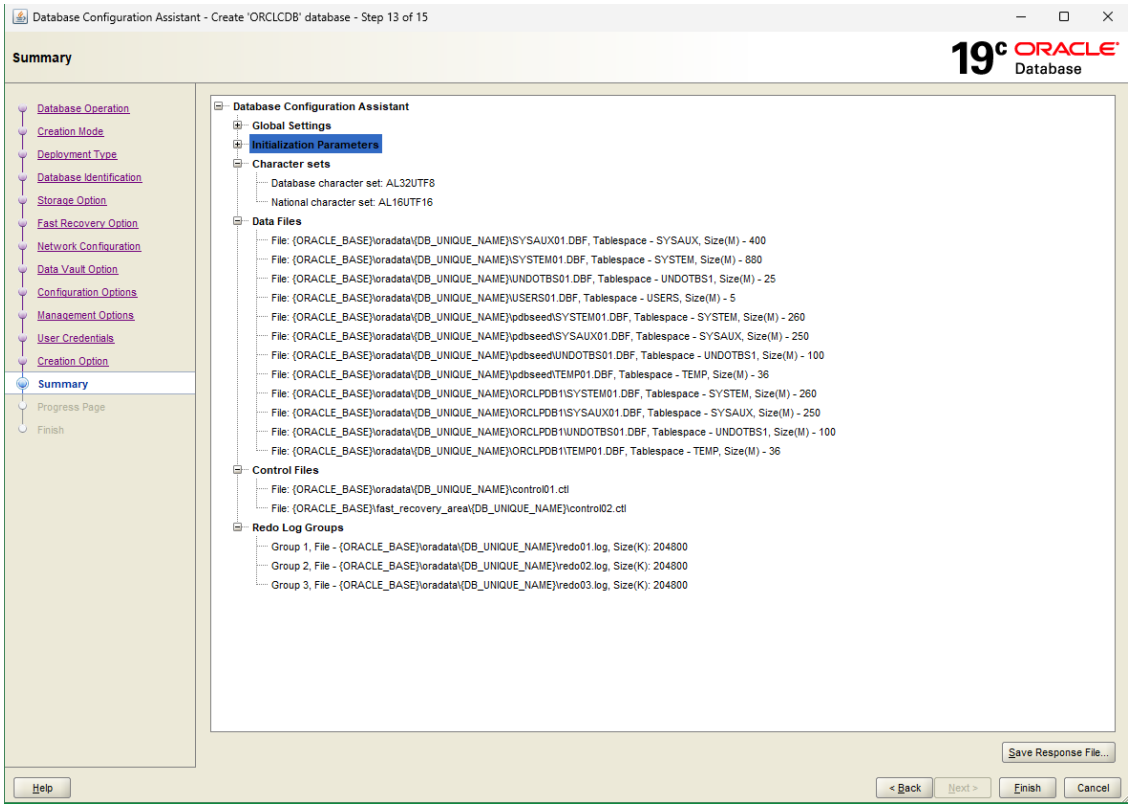


Figure 26. DBCA summary — PDB and CDB datafiles, two control-file copies, three redo groups.

22. Review all values, then click Finish to create ORCLCDB and ORCLPDB1.

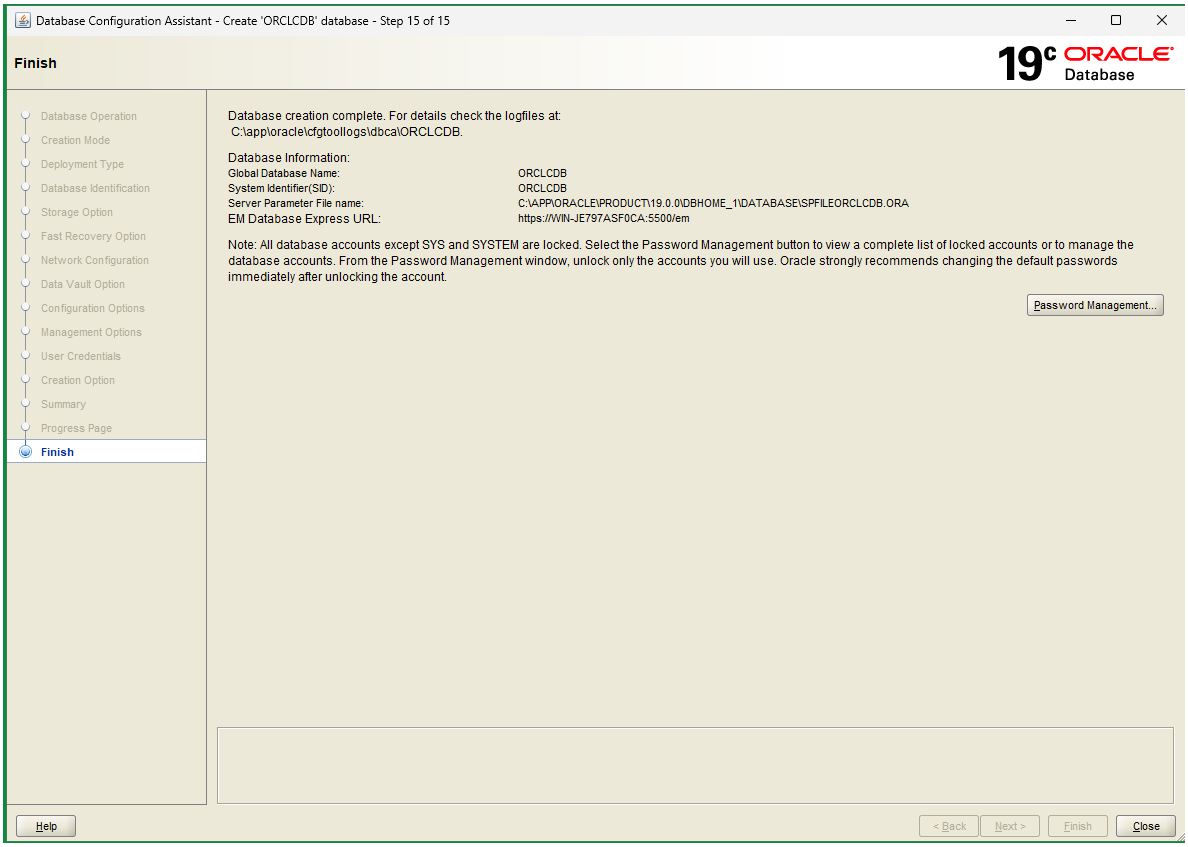


Figure 27. DBCA finish — Database creation complete; instance ORCLCDB and EM Express URL.

DBCA log directory recorded: C:\app\oracle\cfgtoollogs\dbca\ORCLCDB. The displayed EM Express URL uses the server hostname and HTTPS port 5500.

6. Post-installation verification (recorded as passed)

6.1 SQL*Plus local SYSDBA; database and PDB status

In an Administrator PowerShell session, set the SID only for that session and connect locally. Use SQL*Plus commands only at the SQL> prompt.

```
$env:ORACLE_SID = 'ORCLCDB'
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe' / as sysdba
SELECT name, open_mode, log_mode, cdb FROM v$database;
SHOW PDBS;
```

Query / item	Recorded result
v$database	ORCLCDB; READ WRITE; ARCHIVELOG; CDB=YES
PDB$SEED	READ ONLY; RESTRICTED=NO
ORCLPDB1	READ WRITE; RESTRICTED=NO
Local OS authentication	CONNECT / AS SYSDBA succeeded

6.2 Listener, service registration and TCP connection

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\lsnrctl.exe' status LISTENER
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\lsnrctl.exe' services LISTENER
Test-NetConnection -ComputerName 127.0.0.1 -Port 1521
```

Observed: LISTENER running on host WIN-JE797ASF0CA TCP 1521, with ORCLCDB and orclpdb1 services READY; local TCP connection succeeded. EM Express was advertised at TCPS port 5500.

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe' /nolog
-- At SQL> prompt:
CONNECT SYSTEM@"127.0.0.1:1521/orclpdb1"
-- When prompted, enter the admin password privately.
SHOW CON_NAME;
SHOW USER;
```

Recorded output: Connected; CON_NAME=ORCLPDB1; USER=SYSTEM. This established actual Oracle Net authentication, not just TNSPING reachability.

6.3 Windows services and autostart

```
Get-CimInstance Win32_Service | Where-Object Name -Match 'Oracle' |
 Select-Object Name,State,StartMode,StartName | Format-Table -AutoSize
Get-ItemProperty 'HKLM:\SOFTWARE\ORACLE\KEY_OraDB19Home1' |
 Select-Object ORA_ORCLCDB_AUTOSTART,ORA_ORCLCDB_SHUTDOWN,ORA_ORCLCDB_SHUTDOWNTYPE
```

Service / setting	Recorded value
OracleServiceORCLCDB	Running; Automatic; NT SERVICE\OracleServiceORCLCDB
OracleOraDB19Home1TNSListener	Running; Automatic
OracleRemExecServiceV2	Running; Manual
OracleJobSchedulerORCLCDB	Stopped; Disabled (no immediate issue)
OracleVssWriterORCLCDB	Stopped; Automatic (only relevant for VSS workflow)
ORA_ORCLCDB_AUTOSTART	TRUE
ORA_ORCLCDB_SHUTDOWN	TRUE
ORA_ORCLCDB_SHUTDOWNTYPE	immediate

6.4 Save PDB open state

```
-- From a SYSDBA session in CDB$ROOT:
ALTER PLUGGABLE DATABASE ORCLPDB1 SAVE STATE;
SELECT con_name, state, restricted FROM dba_pdb_saved_states;
```

Recorded result: ORCLPDB1 / OPEN / NO. An actual Windows reboot and revalidation remains a recommended but not yet executed test.

6.5 Application tablespace already created

```
-- Use from ORCLPDB1 (not CDB$ROOT)
CREATE TABLESPACE APP_DATA
 DATAFILE 'C:\app\oracle\oradata\ORCLCDB\ORCLPDB1\app_data01.dbf'
 SIZE 100M AUTOEXTEND ON NEXT 50M MAXSIZE 1G
 EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
```

Executed successfully. APP_DATA is ONLINE and PERMANENT, datafile initial size 100 MB, AUTOEXTENSIBLE=YES, maximum 1,024 MB. Do not execute CREATE TABLESPACE again on the existing server.

```
SELECT tablespace_name,status,contents FROM dba_tablespaces
 WHERE tablespace_name='APP_DATA';
SELECT file_name, bytes/1024/1024 size_mb, autoextensible,
      maxbytes/1024/1024 max_mb
FROM dba_data_files WHERE tablespace_name='APP_DATA';
```

7. Create sample data and grant table access (TO EXECUTE)

EXECUTION STATUS The user paused testing immediately after APP_DATA was created. Everything in Sections 7–9 is a documented future procedure, not a completed operation on this server.

Oracle terminology: ORCLCDB is the CDB, ORCLPDB1 is the existing sample/application database boundary, and each database user owns a schema. You do not need another Windows database service or CDB to make sample tables. “Add a user to a table” means grant another user privileges on that table.

7.1 Create an application owner in ORCLPDB1

23. Connect locally as SYSDBA and switch to ORCLPDB1; confirm SHOW CON_NAME displays ORCLPDB1.

```
$env:ORACLE_SID = 'ORCLCDB'
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\sqlplus.exe' / as sysdba
ALTER SESSION SET CONTAINER=ORCLPDB1;
SHOW CON_NAME;
SELECT username FROM dba_users WHERE username IN ('APP_USER','APP_READER','APP_IMPORT');
```

Run the following CREATE USER only if APP_USER does not already exist. SQL*Plus ACCEPT hides the password as it is entered. Use a unique password that is not reused for administration.

```
SET VERIFY OFF
SET ECHO OFF
ACCEPT app_pwd CHAR PROMPT 'New APP_USER password: ' HIDE
CREATE USER APP_USER IDENTIFIED BY "&app_pwd"
```

```
 DEFAULT TABLESPACE APP_DATA
 TEMPORARY TABLESPACE TEMP
 QUOTA 200M ON APP_DATA ACCOUNT UNLOCK;
UNDEFINE app_pwd
GRANT CREATE SESSION, CREATE TABLE TO APP_USER;
```

Grant storage quota on APP_DATA, not UNLIMITED TABLESPACE. No DBA role or CREATE ANY TABLE is required.

7.2 Connect as APP_USER and create sample table

```
-- In SQL*Plus, enter the password only when prompted:
CONNECT APP_USER@"127.0.0.1:1521/orclpdb1"
SHOW USER;
SHOW CON_NAME;
CREATE TABLE CUSTOMERS (
  CUSTOMER_ID NUMBER GENERATED BY DEFAULT ON NULL AS IDENTITY,
  CUSTOMER_NAME VARCHAR2(100 CHAR) NOT NULL,
  EMAIL VARCHAR2(160 CHAR) NOT NULL,
  CREATED_AT TIMESTAMP DEFAULT SYSTIMESTAMP NOT NULL,
  CONSTRAINT PK_CUSTOMERS PRIMARY KEY (CUSTOMER_ID),
  CONSTRAINT UQ_CUSTOMERS_EMAIL UNIQUE (EMAIL)
) TABLESPACE APP_DATA;
```

7.3 Insert and verify sample data

```
INSERT INTO CUSTOMERS (CUSTOMER_NAME,EMAIL) VALUES ('Asha Mehta','asha@example.com');
INSERT INTO CUSTOMERS (CUSTOMER_NAME,EMAIL) VALUES ('Rahul Iyer','rahul@example.com');
INSERT INTO CUSTOMERS (CUSTOMER_NAME,EMAIL) VALUES ('Neha Rao','neha@example.com');
COMMIT;
SELECT CUSTOMER_ID,CUSTOMER_NAME,EMAIL FROM CUSTOMERS ORDER BY CUSTOMER_ID;
SELECT COUNT(*) AS CUSTOMER_COUNT FROM CUSTOMERS;
```

Expected: three rows, CUSTOMER_COUNT=3. The names and example.com addresses are fictional demonstration data. Oracle identity values are generated automatically.

7.4 Add a second user and authorize access to the table

As SYSDBA, switch back to ORCLPDB1, create a login-only read user and grant only SELECT on APP_USER.CUSTOMERS.

```
CONNECT / AS SYSDBA
ALTER SESSION SET CONTAINER=ORCLPDB1;
SET VERIFY OFF
SET ECHO OFF
ACCEPT reader_pwd CHAR PROMPT 'New APP_READER password: ' HIDE
CREATE USER APP_READER IDENTIFIED BY "&reader_pwd"
 DEFAULT TABLESPACE APP_DATA
 TEMPORARY TABLESPACE TEMP ACCOUNT UNLOCK;
UNDEFINE reader_pwd
GRANT CREATE SESSION TO APP_READER;
GRANT SELECT ON APP_USER.CUSTOMERS TO APP_READER;
```

No APP_DATA quota is needed for APP_READER because it is not creating segments. Grant INSERT, UPDATE or DELETE only if a specific application requirement justifies write access.

7.5 Verify owner and second-user access

```
CONNECT APP_READER@"127.0.0.1:1521/orclpdb1"
SHOW USER;
SELECT CUSTOMER_ID,CUSTOMER_NAME,EMAIL
  FROM APP_USER.CUSTOMERS ORDER BY CUSTOMER_ID;
```

Expected: APP_READER can SELECT the same three rows. If an attempt to INSERT is made as APP_READER, it should fail unless DML permissions were explicitly granted. Do not use SYS or SYSTEM for normal application traffic.

7.6 Verify user metadata (administrator session)

```
CONNECT / AS SYSDBA
ALTER SESSION SET CONTAINER=ORCLPDB1;
SELECT username,account_status,default_tablespace
 FROM dba_users WHERE username IN ('APP_USER','APP_READER');
SELECT owner,table_name FROM dba_tables
 WHERE owner='APP_USER' AND table_name='CUSTOMERS';
SELECT grantee,privilege FROM dba_tab_privs
 WHERE owner='APP_USER' AND table_name='CUSTOMERS';
```

8. RMAN backup of the whole CDB (TO EXECUTE)

8.1 Backup prerequisites and safety checks

The database is in ARCHIVELOG mode, so online RMAN backup is supported. Connect RMAN to the CDB root: a whole-CDB backup includes the root, seed and all PDB datafiles. Back up archived redo logs as well. Do not confuse a Data Pump dump with a recoverable physical database backup.

- Create a backup destination with enough free space. This procedure uses C:\OracleBackups\ORCLCDB for a lab; prefer a dedicated backup disk or off-host repository in production.
- Confirm Windows service identity NT SERVICE\OracleServiceORCLCDB has Modify access to the backup directory; restricting other users is recommended.
- Do not place the only backup on the same disk as the database in a real environment.
- Confirm no process is out of FRA space. The configured FRA quota is only 12,732 MB, so avoid storing an unmonitored large backup there.

```
# Run from elevated PowerShell; verify the folder and ACL afterward
New-Item -ItemType Directory -Force 'C:\OracleBackups\ORCLCDB'
icacls 'C:\OracleBackups\ORCLCDB' /grant 'NT SERVICE\OracleServiceORCLCDB:(OI)(CI)M'
icacls 'C:\OracleBackups\ORCLCDB'
Get-PSDrive C
```

ACCESS CONTROL This ACL command is a starting point for the recorded virtual service account. If Windows cannot resolve the account or RMAN reports a file-creation error, inspect the actual service identity and permissions rather than granting Everyone Full Control.

8.2 Start RMAN and configure backup output

Close any SQL*Plus process, open Administrator PowerShell and connect RMAN to the local CDB as SYSDBA via operating-system authentication.

```
$env:ORACLE_SID = 'ORCLCDB'
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\rman.exe' TARGET /
```

At the RMAN> prompt:

```
SHOW ALL;
CONFIGURE CONTROLFILE AUTOBACKUP ON;
CONFIGURE CHANNEL DEVICE TYPE DISK
 FORMAT 'C:\OracleBackups\ORCLCDB\bk_%U.bkp';
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK
 TO 'C:\OracleBackups\ORCLCDB\cf_%F.bkp';
```

These RMAN CONFIGURE settings persist. If preferred, replace C: with the fully qualified path of a separately mounted backup volume before running the commands.

8.3 Take an online full backup with archived logs

```
BACKUP DATABASE PLUS ARCHIVELOG;
LIST BACKUP SUMMARY;
LIST BACKUP OF DATABASE;
LIST BACKUP OF ARCHIVELOG ALL;
```

BACKUP DATABASE PLUS ARCHIVELOG is the Oracle-documented single-command way to back up database files and required archived logs. Do not manually delete archive logs to make space. Configure an appropriate retention and deletion policy only after confirming the backup regime.

8.4 Verify physical backups without restoring

```
CROSSCHECK BACKUP;
RESTORE DATABASE VALIDATE;
LIST BACKUP SUMMARY;
EXIT;
```

RESTORE DATABASE VALIDATE reads backup metadata and validates the files needed for restore but does not overwrite the live database. A real disaster-recovery test still requires an isolated restore exercise.

8.5 Validate FRA consumption from SQL*Plus

```
CONNECT / AS SYSDBA
SELECT name,space_limit/1024/1024 AS limit_mb,
    space_used/1024/1024 AS used_mb,
    space_reclaimable/1024/1024 AS reclaimable_mb
FROM v$recovery_file_dest;
```

Expected backup acceptance: RMAN command completed successfully, LIST BACKUP shows backup sets, RESTORE DATABASE VALIDATE completes, and backup files exist in the backup destination. Preserve copies outside the VM.

9. Export and import the sample schema (TO EXECUTE)

9.1 Prepare a Data Pump directory

Data Pump dump and log files are written by Oracle server processes to a server-side Windows path, not to the local client working directory. A database DIRECTORY object maps an Oracle name to that OS path.

```
# Administrator PowerShell on the database server
New-Item -ItemType Directory -Force 'C:\OracleExports\ORCLPDB1'
```

```
icacls 'C:\OracleExports\ORCLPDB1' /grant 'NT SERVICE\OracleServiceORCLCDB:(OI)(CI)M'
```

In a SYSDBA SQL*Plus session, switch to ORCLPDB1 and create the DIRECTORY object. Do not create it in CDB$ROOT for this PDB export.

```
CONNECT / AS SYSDBA
ALTER SESSION SET CONTAINER=ORCLPDB1;
CREATE OR REPLACE DIRECTORY APP_DPUMP AS 'C:\OracleExports\ORCLPDB1';
GRANT READ, WRITE ON DIRECTORY APP_DPUMP TO APP_USER;
SELECT directory_name,directory_path FROM dba_directories
 WHERE directory_name='APP_DPUMP';
```

9.2 Export APP_USER schema with expdp

Exit SQL*Plus. Run the executable in PowerShell; enter the APP_USER password at its prompt, not as a visible command-line argument.

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\expdp.exe' `
 'APP_USER@127.0.0.1:1521/orclpdb1' `
 'SCHEMAS=APP_USER' 'DIRECTORY=APP_DPUMP' `
 'DUMPFILE=app_user_01.dmp' 'LOGFILE=app_user_export_01.log' `
 'LOGTIME=ALL' 'METRICS=YES'
```

If SQL*Plus or Data Pump misparses a password containing @, use a properly quoted password at the interactive prompt. Prefer unique application credentials without quoting-sensitive characters for the initial lab, while still enforcing sufficient complexity. Do not place passwords in parfiles.

Verify C:\OracleExports\ORCLPDB1\app_user_01.dmp and app_user_export_01.log exist; inspect the log for “successfully completed”. The dump is logical schema transport, not a substitute for an RMAN whole-database backup.

9.3 Create an empty target user for import

Use a separate schema APP_IMPORT so the exported APP_USER data can be imported without overwriting the source objects.

```
-- SQL*Plus: run as SYSDBA inside ORCLPDB1
CONNECT / AS SYSDBA
ALTER SESSION SET CONTAINER=ORCLPDB1;
SET VERIFY OFF
SET ECHO OFF
ACCEPT import_pwd CHAR PROMPT 'New APP_IMPORT password: ' HIDE
CREATE USER APP_IMPORT IDENTIFIED BY "&import_pwd"
 DEFAULT TABLESPACE APP_DATA TEMPORARY TABLESPACE TEMP
 QUOTA 200M ON APP_DATA ACCOUNT UNLOCK;
UNDEFINE import_pwd
GRANT CREATE SESSION, CREATE TABLE TO APP_IMPORT;
GRANT READ, WRITE ON DIRECTORY APP_DPUMP TO SYSTEM;
```

APP_IMPORT has a storage quota but no blanket administrative privileges. The SYS/SYSTEM-level account executes the remap import; do not grant DBA to APP_IMPORT.

9.4 Import into APP_IMPORT using REMAP_SCHEMA

Exit SQL*Plus. In PowerShell invoke impdp as SYSTEM and provide its password privately when prompted. The export must have completed successfully first.

```
& 'C:\app\oracle\product\19.0.0\dbhome_1\bin\impdp.exe' `
 'SYSTEM@127.0.0.1:1521/orclpdb1' `
 'SCHEMAS=APP_USER' 'REMAP_SCHEMA=APP_USER:APP_IMPORT' `
 'DIRECTORY=APP_DPUMP' 'DUMPFILE=app_user_01.dmp' `
 'LOGFILE=app_user_import_01.log' 'LOGTIME=ALL' 'METRICS=YES'
```

SYSTEM must be authorized for schema import/remapping. A SYSDBA account should be reserved for necessary administrative tasks. Do not import into SYS or SYSTEM. For an existing target table, decide explicitly whether to SKIP, APPEND, TRUNCATE or REPLACE before specifying TABLE_EXISTS_ACTION.

9.5 Validate the imported table and compare row counts

```
-- SQL*Plus: authenticate as APP_IMPORT to ORCLPDB1
CONNECT APP_IMPORT@"127.0.0.1:1521/orclpdb1"
SELECT COUNT(*) AS IMPORTED_CUSTOMERS FROM CUSTOMERS;
SELECT CUSTOMER_ID,CUSTOMER_NAME,EMAIL FROM CUSTOMERS ORDER BY CUSTOMER_ID;
-- Expected sample: three rows, if Section 7 inserted three rows.
```

Compare schema object lists and row counts against APP_USER. For production, reconcile object validity, constraints, grants, indexes, statistics, application sequences and application-specific dependencies.

10. Troubleshooting and operational cautions

10.1 Resolved SQL*Plus ORA-12543: password contains @

The lab originally showed ORA-12504 for an incorrectly parsed connection and ORA-12543 even though listener status, handler status, Test-NetConnection and TNSPING succeeded. The client sqlnet.log repeatedly displayed HOST=0.0.0.123 and an empty SERVICE_NAME. The shared administrator password contained @, and enclosing the password in double quotes at the interactive prompt successfully established the SYSTEM connection to ORCLPDB1.

```
-- At SQL> in SQL*Plus 19c:
CONNECT SYSTEM@"127.0.0.1:1521/orclpdb1"
-- At the password prompt, supply the existing secret enclosed in double quotes.
-- Do not type the password into copied documentation or chat.
```

Current sqlnet.ora was changed during troubleshooting to NAMES.DIRECTORY_PATH=(EZCONNECT,TNSNAMES); SQLNET.AUTHENTICATION_SERVICES=(NTS) was retained. The decisive successful change was quoting the password, not changing the listener or firewall.

10.2 Quick fault matrix

Symptom	Check / response
sqlplus not found	Use full path under Oracle Home\bin, or configure session PATH only.
ORA-12541	Check OracleOraDB19Home1TNSListener service and lsnrctl status.
ORA-12514	Check service name ORCLPDB1, SHOW PDBS and lsnrctl services.
ORA-12504	Verify an explicit SERVICE_NAME or quoted Easy Connect string.
ORA-12543 with HOST=0.0.0.123	Check credentials with @ and SQL*Plus parsing; quote password at prompt.
ORA-01017	Recheck username, password, target PDB and account state.
ORA-01950	Grant intended owner a quota on APP_DATA.
Data Pump ORA-39070 / UTL_FILE errors	Validate OS folder, DIRECTORY object and service-account ACL.

RMAN file create errors	Check target directory, disk space, service-account ACL and RMAN channel format.
ARCHIVELOG / FRA full	Review v$recovery_file_dest, backup copies and compliant log deletion policy.

10.3 Hardening / future maintenance

- Obtain and apply Oracle Database 19c RU 19.27 or later (plus a compatible OPatch) before using Windows Server 2025 as a supported deployment; follow the specific bundle patch README and run datapatch after patching an existing database.
- Replace the shared initial administrative password with distinct SYS, SYSTEM and PDBADMIN passwords; keep secrets out of scripts and logs.
- Restrict listener TCP 1521 and EM Express TCPS 5500 to authorized lab/admin networks.
- Do not grant the DBA role to applications; keep APP_USER, APP_READER and APP_IMPORT scoped to their use cases.
- Configure monitored backup retention, off-host backup copies, restore testing and FRA capacity thresholds.
- Reboot in a planned maintenance window, then repeat startup, SHOW PDBS, listener and PDB login verification.

11. Acceptance checklist and handover

Milestone	Evidence / acceptance condition	Recorded status
Previous 26ai removed	No Free services; C:\app\oracle\26ai removed	PASS
Oracle Home installed	19.3 SQL*Plus and DBCA available; Oracle registry correct	PASS
CDB/PDB created	ORCLCDB READ WRITE, ORCLPDB1 READ WRITE	PASS
ARCHIVELOG configured	v$database.log_mode=ARCHIVELOG	PASS
Listener and network login	ORCLPDB1 READY and SYSTEM connected via TCP	PASS
Windows services / saved state	Autostart TRUE; ORCLPDB1 saved OPEN	PASS
Application tablespace	APP_DATA ONLINE, 100 MB / 1 GB max	PASS
Sample schema / second user	Create APP_USER + CUSTOMERS + APP_READER; test SELECT	NOT EXECUTED
RMAN backup / validate	Backup CDB + archivelogs; RESTORE DATABASE VALIDATE	NOT EXECUTED
Data Pump export/import	APP_USER to APP_IMPORT; count 3 sample rows	NOT EXECUTED
Supported OS/RU combination	Oracle 19c RU >=19.27 on Windows Server 2025	NOT MET
Reboot and restore drill	Post-reboot smoke / independent recovery exercise	NOT EXECUTED

Handover note: Do not interpret successful installation or connection tests as Oracle certification. For any customer deployment, close the release-update, licensing, recovery, network-security and operational acceptance gaps before production sign-off.

12. Official reference links

- Installation Guide, 19c for Microsoft Windows — https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/
- Windows Server support / minimum release updates — https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/operating-system-checklist-oracle-database-installation-microsoft-windows.html

- Install Oracle Database software — https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/installing-oracle-database.html
- Oracle Database Administrator's Guide / DBCA overview — https://docs.oracle.com/en/database/oracle/oracle-database/19/admqs/installing-oracle-database-creating-database.html
- Backup and Recovery User Guide — https://docs.oracle.com/en/database/oracle/oracle-database/19/bradv/backing-up-database.html
- RMAN quick start — https://docs.oracle.com/en/database/oracle/oracle-database/19/bradv/getting-started-rman.html
- Oracle Data Pump documentation — https://docs.oracle.com/en/database/oracle/oracle-database/19/sutil/
- SQL GRANT reference — https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/GRANT.html
- Oracle Software Delivery Cloud — https://edelivery.oracle.com/osdc/faces/SoftwareDelivery
- My Oracle Support — https://support.oracle.com/